

DecodeLabs Data Analytics Internship

Project 1: Data Cleaning & Preparation

Submitted By

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1. Introduction

Objective

The objective of this project is to clean and prepare a raw dataset for analysis by identifying missing values, removing duplicate records, correcting inconsistent data formats, and ensuring overall data quality. A clean dataset is essential for reliable analysis, visualization, and decision-making.

2. Dataset Information

The dataset contains customer order information from an online retail business.

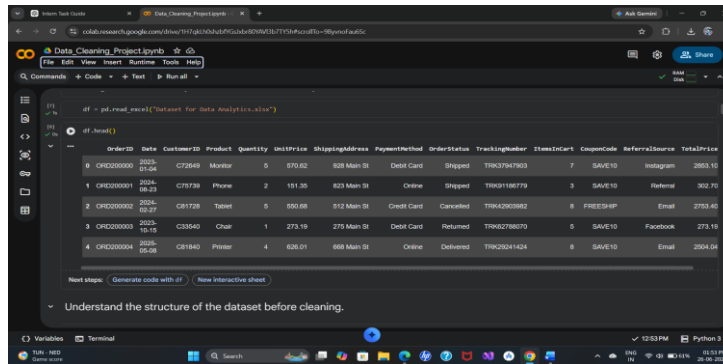
Dataset Features

- OrderID
- Date
- CustomerID
- Product
- Quantity
- UnitPrice
- ShippingAddress
- PaymentMethod
- OrderStatus
- TrackingNumber
- ItemsInCart
- CouponCode
- ReferralSource

- TotalPrice

Initial Dataset Summary

- Total Rows: 1200
- Total Columns: 14
- File Format: Excel (.xlsx)



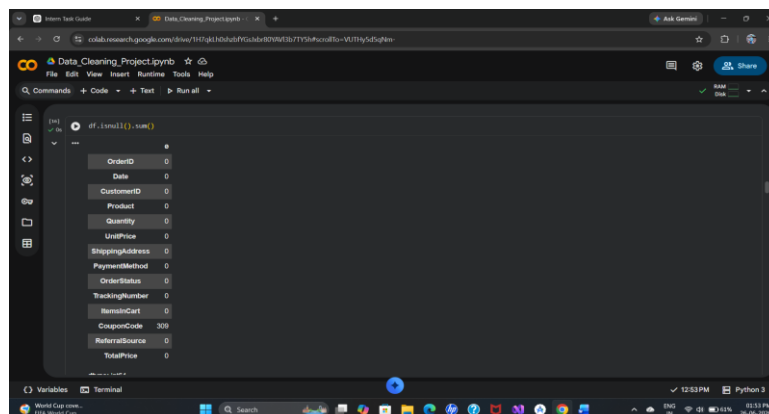
	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
0	ORD000000	2023-01-04	C72649	Monitor	5	570.62	828 Main St	Debit Card	Shipped	TRN37947903	7	SAVE10	Instagram	2853.10
1	ORD000001	2024-08-23	C73730	Phone	2	181.36	833 Main St	Online	Shipped	TRN31188779	3	SAVE10	Referral	362.72
2	ORD000002	2024-02-27	C81728	Tablet	5	590.88	512 Main St	Credit Card	Cancelled	TRN42903982	8	FREE54P	Email	2753.40
3	ORD000003	2025-10-19	C33540	Chair	1	273.19	279 Main St	Debit Card	Returned	TRN652788070	5	SAVE10	Facebook	273.19
4	ORD000004	2025-05-08	C81840	Printer	4	626.51	688 Main St	Online	Delivered	TRN26241424	8	SAVE10	Email	2504.04

3. Data Cleaning Process

The following data cleaning steps were performed:

✓ Missing Value Analysis

- Checked missing values using Pandas.
- Filled missing numerical values using the median.
- Filled missing categorical values with "Unknown".



Column	Count
OrderID	0
Date	0
CustomerID	0
Product	0
Quantity	0
UnitPrice	0
ShippingAddress	0
PaymentMethod	0
OrderStatus	0
TrackingNumber	0
ItemsInCart	0
CouponCode	309
ReferralSource	0
TotalPrice	0

✓ Duplicate Record Removal

- Identified duplicate rows.
- Removed duplicate records.
- Verified that no duplicate rows remained.

- The cleaned dataset is suitable for further business analysis.
 - Basic exploratory analysis revealed customer purchasing trends and payment preferences.
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6. Tools & Technologies

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Microsoft Excel
 - Jupyter Notebook
 - GitHub
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7. Conclusion

This project demonstrates the complete process of transforming raw data into a structured, reliable, and analysis-ready dataset. The cleaned data can now be used for dashboards, reporting, predictive analytics, and business decision-making. The project strengthened practical skills in data preprocessing, validation, and exploratory analysis.

8. Project Deliverables

- ✓ Raw Dataset
 - ✓ Cleaned Dataset
 - ✓ Jupyter Notebook
 - ✓ Project Report
 - ✓ GitHub Repository
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GitHub Repository

<https://github.com/ashwini-s2004/DecodeLabs-Data-Cleaning-Project>